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**Deriverable 6. Key Conlusions**

**¨Automatic Classification of Hydroxychloroquine-Induced Retinal Toxicity using OCT and Deep Learning¨**

**Team No. 45**

Natalia Nevarez Tinoco – A01566204

Alejandro Caamaño Granados – A00843392

Diana Jimena López Nájera – A01024913

**Faculty Team:**

Dra. Grettel Barceló Alonso – Lead Professor and Project Advisor

Dr. Luis Eduardo Falcón Morales – National Director, MNA

Mtra. Verónica Sandra Guzmán de Valle – Assistant Professor

**Sponsor:**

Dr. Alejandro Rodríguez García - Hospital Zambrano Hellion

# Index



Patients on long-term hydroxychloroquine (HCQ) therapy face a slow, often symptomless threat: the drug can accumulate in the retina and cause irreversible macular damage. Spectral-domain optical coherence tomography (SD-OCT) can reveal this damage early, but reliable interpretation depends on scarce specialist time, and the toxicity signal is subtle. This project asked a focused question: **can a deep learning model screen SD-OCT images and flag likely HCQ toxicity, so that specialist attention is directed where it matters most?** What follows is the honest answer we reached after working through all six phases of the CRISP-ML(Q) lifecycle (Studer et al., 2021), the problem, what the data taught us, the models we built, the recommendation, and a clear accounting of cost, benefit, and risk.

# 1. The Problem

The administration of hydroxychloroquine (HCQ) remains a cornerstone in the long-term management of systemic autoimmune conditions, such as lupus erythematosus and rheumatoid arthritis. This therapeutic reliance is deeply rooted in clinical history; its predecessor, chloroquine, was developed in 1939 to treat malaria during World War II, where it unexpectedly demonstrated significant secondary benefits for patients suffering from autoimmune disorders. In 1955, its analog, hydroxychloroquine, was synthesized and rapidly replaced chloroquine due to its superior safety and tolerability profile. Nevertheless, despite its systemic benefits, prolonged accumulation of the drug carries a critical, well-documented risk of inducing irreversible retinal toxicity.

The current clinical workflow at Hospital Zambrano Hellion relies on manual inspection of Spectral-Domain Optical Coherence Tomography (SD-OCT) images. These B-scans provide high-resolution details of the retinal anatomy, allowing clinicians to identify subtle structural changes. In cases of HCQ toxicity, characteristic findings include the loss of the parafoveal ellipsoid zone, often visualized as a "flying saucer" sign. Consequently, this manual review process introduces high inter-observer variability, imposes a significant operational burden on ophthalmologists, and creates bottlenecks that can lead to hazardous delays in diagnosis.

The justification for this project lies in the transformative potential of artificial intelligence to augment clinical diagnostics. While global studies have demonstrated the efficacy of deep learning in identifying HCQ toxicity markers, our project is unique in its application to a local Mexican population. By creating an automated tool that standardizes the classification of OCT B-scans, we aim to provide an objective second opinion that increases diagnostic accuracy, reduces the time burden on clinicians, and facilitates early clinical intervention for patients in our local healthcare ecosystem.

## Business understanding

The core problem is that manual interpretation of OCT imaging is a repetitive and cognitively demanding task that creates a bottleneck in ophthalmological care. The clinical objective is to bridge this gap by developing a robust deep learning model that classifies images as either containing signs of toxicity or being healthy. By automating this binary classification, the project aims to minimize the subjectivity of human assessment and provide a high-precision tool that supports specialists in making informed decisions about patient safety and treatment continuity.

Our general objective is to validate that a deep learning framework can reliably identify retinal toxicity, supported by specific goals such as building a high-quality labeled dataset, comparing scratch-trained models against advanced transfer learning architectures, and establishing a prototype for clinical validation. Key questions center on how to optimize the model's performance despite severe data imbalances and how to ensure the system remains interpretable for clinical use. Ultimately, this tool is intended to empower the ophthalmology team at Hospital Zambrano Hellion to improve patient outcomes through faster and more reliable screening.

The primary stakeholders include our academic team—Natalia Nevarez, Alejandro Caamaño, and Jimena López—alongside our project advisor, Dra. Grettel Barceló and our medical sponsor Dr. Alejandro Rodríguez García. Each stakeholder plays a vital role in ensuring the solution aligns with both technical excellence and real-world medical practice. By bringing together AI expertise and specialized ophthalmological knowledge, we ensure that the project is not merely an academic exercise but a practical contribution to clinical infrastructure that addresses a known medical need.

# 2. What the Data Told Us

The exploratory analysis did not just describe the dataset; it redefined the problem. Three findings shaped everything that followed.

**Severe class imbalance built on very few patients**. The dataset contained roughly 179 "No toxicity" images against only 15 "Toxicity" images, and those 15 positive images came from just 5 distinct patients. The effective positive sample size is therefore tiny, which is the single hardest constraint of the project and the reason ordinary accuracy figures are meaningless here.

**Patient-level leakage is the central methodological trap**. Because multiple images can come from one patient, a naïve random split can place images of the same eye in both training and test sets, letting the model "recognize the patient" rather than learn the disease. We treated this as non-negotiable: all images from a patient must stay in the same fold, enforced with **StratifiedGroupKFold** cross-validation and a held-out test set that deliberately isolates positive patients.

**Confounders that look like signal but are not.** We found that file format, file size, and image resolution correlated with the class label, not because of pathology, but because positive and negative images were acquired under different conditions (“storage fingerprints”). Separately, toxicity-positive patients tended to be older and more likely to have cataracts, which physically degrade OCT image quality (van Velthoven et al., 2006). Both effects can inflate apparent performance and were flagged for exclusion or careful handling rather than exploited.

# 3. Models Built and Why We Chose the Final One

We evaluated a progression of approaches under consistent, leakage-safe cross-validation: classical baselines, transfer-learning with frozen pretrained features, and full fine-tuning of a convolutional backbone, with hyperparameter search via Optuna (Akiba et al., 2019) and Grad-CAM for interpretability (Selvaraju et al., 2017). Three lessons drove the final selection.

Fine-tuning beat frozen features at this scale. An earlier conclusion favoring frozen features was reversed once we measured carefully: full fine-tuning delivered substantially better precision-recall performance and became our final model. Threshold selection mattered more than model selection. At the default of 0.5 cutoff, classical models collapsed to zero recall; they simply never predicted the rare class. Sweeping the decision threshold down to ~0.26 recovered a recall of 0.933, turning an unstable model into a viable screen. Finally, we report honest metrics, not flattering ones. Development-only metrics suggested an attractive ~0.93 through hard-voting, but that figure is a mirage of the small evaluation set. Our headline is the full-data out-of-fold result: a PR-AUC of approximately 0.52 with a recall near 0.73. We chose PR-AUC over ROC-AUC because it is the appropriate measure under sever imbalance, and we optimized toward F2 rather than F1 because F2 weights recall more heavily; the right emphasis when missed casses are the expensive error.

| ~0.52 | ~0.73 | F2 |
|---|---|---|
| PR-AUC (full-data out-of-fold — honest headline) | Recall at operating point (0.933 via threshold tuning) | Optimization target (prioritizes avoiding misses) |

Interpretability gave us our most important caution. Grad-CAM showed the model concentrating on the central foveal region, whereas the clinical literature locates early HCQ toxicity in the parafoveal / pericentral outer retinal layers (Marmor et al., 2016). This misalignment is not a finding that should be hidden, it is a documented finding and a concrete benchmark for future iterations, telling us the model had not yet learned the clinically correct feature location.

# 4. Key Recommendations for Implementation

Our honest read of the evaluation leads to one overriding recommendation: staged deployment; the model is not yet ready for clinical use. Promising results from five positive patients are not a foundation for decisions about the vision. The path forward is sequenced deliberately.

- **Make the data deployment ready now**. The highest value-action available today requires no new patients: standardize file formats, naming conventions, and patient-level identifiers across the existing workflow. This eliminates the storage-fingerprint confounders and ensures that every image accumulating from here forward is clean and immediately usable the moment the dataset reaches sufficient scale.
- **Grow the distinct-patient base as background work**. More images from the same patients are easy to obtain but do little for generalization; new distinct patients are what the model needs, and they accumulate slowly given the rarity of the condition. This is framed as a steady background process, not an urgent ask.
- **Adopt a cloud platform aligned with the existing workflow**. We recommend Google Cloud (Vertex AI + the Medical Imaging Suite) for continuity with the current Colab/Drive environment and its native handling of DICOM medical imaging (Google Cloud, 2026).
- **Treat the Grad-CAM benchmark as a release gate.** Before any clinical pilot, future iterations should demonstrate attention to the clinically correct parafoveal region.

# 5. Cost-Benefit Analysis

All figures are rational planning estimates for an academic proof-of-concept, not audited expenditures. Converted at ~17.2 MXN per USD (June 2026). Team labor is in-kind (graduate student effort) and shown at a notional value to convey the true resource investment.

**Costs incurred to date, by CRISP-ML(Q) phase**

| CRISP-ML(Q) phase | Main cost drivers | USD | MXN |
|---|---|---|---|
| Business & Data Understanding | Sponsor meetings, scoping, EDA; Colab Pro compute, Drive storage | $110 | $1,890 |
| Data Preparation | Cleaning, de-confounding, leakage-safe splitting (labor only, open-source tools) | $0 cash | $0 cash |
| Modeling | Intensive GPU training, Optuna search, fine-tuning (Colab Pro+ compute) | $150 | $2,580 |
| Evaluation | Cross-validation runs, threshold sweeps, Grad-CAM compute | $40 | $690 |
| Software & tooling | Optuna, Grad-CAM, scikit-learn, PyTorch — all open source | $0 | $0 |
| **Direct cash cost to date** | | **$300** | **$5,160** |

**Projected operation & maintenance (annual, post-deployment)**

| Item | Description | USD / year | MXN / year |
|---|---|---|---|
| Vertex AI + Medical Imaging Suite | Managed inference, DICOM storage, model hosting | $200 – $400 | $3,440 – $6,880 |

| | | | |
|---|---|---|---|
| Periodic retraining | Compute as new distinct patients accumulate | $100 – $200 | $1,720 – $3,440 |
| Clinical validation cycles | Specialist time to confirm flagged cases (in-kind) | in-kind | in-kind |
| **Estimated annual run cost** | | **$300 – $600** | **$5,160 – $10,320** |

## Benefits

**Quantifiable**. A specialist SD-OCT reading session in a private hospital context runs roughly $150-$250 USD ($2,580 - $4,300 MXN) per patient. An effective triage layer that confidently clears low-risk scans reduces unnecessary specialist referrals; even modest reductions across a screening population offset the modest annual run cost many times over. More importantly, each true toxicity case caught early avoids the downstream cost of permanent vision loss, a clinical and liability burden for exceeding any compute expense.

**Intangible**. A standardized, reproducible screening protocol across TecSalud facilities; reduced cognitive load on ophthalmologists; a clean, well governed dataset that becomes a durable research asset, and a reusable template for AI-assisted monitoring of other rare retinal conditions.

# 6. Risk and Challenges

We classify the principal risks using the four categories from the course framework.

## Data

- Extreme imbalance and few patients (179 No / Si from 5 patients); the dominant limit on generalization and the reason the model is not yet deployable.
- Storage-fingerprint confounders, such as format, size, and resolution correlate with the label through acquisition differences, risking models that learn the scanner rather than the disease.
- Patient-level confounders. Positive patients skew older with more cataracts, degrading image quality in ways correlated with the label.
- No standardized acquisition protocal across future contributors, threatening data quality as the set grows.

## Attacks

- Adversarial perturbations – small, imperceptible image changes could flip predictions in a clinical setting.
- Data poisoning – unverified incoming images could corrupt future retraining.
- Privacy / model-inversion exposure of patient-identifiable patterns, given the very small positive cohort.

## Testing & Trust

- Grad-CAM misalignment – the model attends to the foveal center, not the parafoveal region where HCQ toxicity actually manifests; trust must not outrun this evidence.
- Over-reliance – clinicians could over-trust a model whose honest PR-AUC is ~0.52; clear communication of limits is essential.
- Distribution shift – performance may drift as the dataset and acquisition conditions evolve.

## Compliance

- Health and data governance – DICOM/PHI handling must meet Mexican regulation (e.g. NOM – 004) (NOM-004-SSA3-2012; DOF, 2012) and institutional ethics requirements.
- Medical-device status – the model is not cleared as a medical device (COFEPRIS); clinical use is premature without an appropriate review (COFEPRIS, 2025).
- Liability - using model output in clinical decisions before validation transfers undue risk to clinicians and the institution.

# 7. Conclusions

This project set out to build a deep learning screen for HCQ-induced macular toxicity, and, across six phases of CRISP-ML(Q), it produced two things of value: a methodologically disciplined prototype and an unusually clear understanding of its own limits. The technical decisions that mattered most were not about architecture but about honesty, enforcing patient-level separation to avoid leakage, choosing PR-AUC and F2 as the metrics that fit a rare, high-cost-of-miss condition, and reporting full-data out-of-fold results rather than the flattering numbers a small evaluation set can produce.

The central conclusion is that performance at this stage is bounded not by modeling choices but by data: the toxicity class rests on too few distinct patients for the model to generalize, and interpretability confirms it has not yet learned the clinically correct parafoveal signal (Marmor et al., 2016). This echoes, at a smaller scale, the sponsor's own prior SD-OCT classification work, where strong results were achieved on a modest, carefully handled image set (Bustamante-Arias et al., 2021). Our recommendation follows directly and is deliberately low-cost: standardize data capture now so that every future scan is immediately usable, grow the distinct-patient base as steady background work, and treat the Grad-CAM benchmark as a gate before any clinical pilot. The model is not yet ready for clinical use, and saying so plainly is the most useful conclusion we can offer a decision-maker.

# 8. Preliminary Findings: Milestone 5 Re-Run with Expanded Si Data

Everything above reflects the pipeline our team developed and validated over the past six weeks; the conclusions of this summary stand on that body of work. In parallel, we have begun a further experiment that we report here honestly even though its analysis is not yet complete.

We re-ran the full Milestone 5 modeling pipeline after expanding the toxicity (Si) class from roughly 15 images (5 patients) to approximately 260 images drawn from **11 distinct patients,** more than four times the positive images and more than double the positive patients. The reasonable expectation was that substantially more positive data would raise PR-AUC, our primary metric. **It did not.** The headline full-data out-of-fold PR-AUC did not improve over the established pipeline; on a like-for-like basis it came down, while generalization to genuinely unseen patients remained poor.

**Expanded-Si run — key results (best model: MobileNetV2)**

| Metric | Established (≤5 Si pts) | Expanded (11 Si pts) |
|---|---|---|
| Development OOF PR-AUC (best model) | ~0.52 | 0.47 |
| Full-data OOF PR-AUC (best model) | ~0.52 | 0.29 |
| Held-out test PR-AUC (unseen patients) | ~0.13 | 0.21 |
| Si recall (full-data OOF) | ~0.73 | 0.77 |

Established-pipeline figures are the full-data OOF / held-out values reported in Sections 3 above (≤5 Si patients). Development-only OOF was confirmed optimistic: across models it averaged ~0.08 PR-AUC higher than the full-data estimate.

### Why the PR-AUC did not rise

**Distinct patients, not image count, govern the metric.** Under patient-grouped cross-validation, PR-AUC measures generalization to patients the model has never seen. Moving from 5 to 11 Si patients is still a very small number of distinct disease presentations, and images from one patient are highly correlated, so the effective sample size grew far less than the raw image count suggests.

**The earlier figure was optimistic.** With only five Si patients, each fold held out roughly one positive patient, making the estimate unstable and easily flattered by chance. Evaluating across eleven patients yields a more trustworthy, and lower, number. Part of the apparent drop is therefore the estimate becoming more honest, not the model becoming worse.

**Generalization still fails on unseen patients.** On the forced held-out test set of truly unseen Si patients, PR-AUC stayed near 0.21 with recall near 0.29. More positive images did not fix the core problem: the model does not yet transfer to new patients.

**The model still attends to the wrong region.** Grad-CAM continues to highlight the central fovea rather than the parafoveal zone where toxicity manifests (Marmor et al., 2016). Feeding more images of a feature the model is not learning correctly does not improve it.

**Added heterogeneity.** The six additional patients bring new ages, cataract status, and acquisition conditions, genuine clinical variability that a model trained on so few patients cannot yet absorb.

**What it means, and why this is preliminary**

This result reinforces rather than overturns the central thesis of the project: **distinct-patient diversity, not image volume, is the binding constraint on performance.** It is direct evidence for the data-readiness recommendation in Section 4, and a caution against trusting any single headline number while patient counts remain this small.

Because these expanded-Si results are recent, they still require deeper analysis before we can draw firm conclusions: per-patient error analysis, a fresh leakage audit on the new identifiers, calibration and threshold behavior on the new patients, and an investigation of whether the six additional patients differ systematically from the original cohort. For that reason this executive summary is built on the validated six-week pipeline, and the expanded run is presented here as an early finding to be understood, not as a new headline result. We will keep iterating to learn how and why these numbers arose. The honest takeaway is that more data helped us see the problem more clearly, not that it solved it, and seeing it clearly is precisely what tells us where to invest next.

# 9. References

Akiba, T., Sano, S., Yanase, T., Ohta, T., & Koyama, M. (2019). Optuna: A next-generation hyperparameter optimization framework. In *Proceedings of the 25th ACM SIGKDD International Conference on Knowledge Discovery & Data Mining* (pp. 2623–2631). Association for Computing Machinery. https://doi.org/10.1145/3292500.3330701

Babic, B., Cohen, I. G., Evgeniou, T., & Gerke, S. (2021). When machine learning goes off the rails. *Harvard Business Review, 99*(1), 76–84. https://hbr.org/2021/01/when-machine-learning-goes-off-the-rails

Bustamante-Arias, A., Cheddad, A., Jimenez-Perez, J. C., & Rodriguez-Garcia, A. (2021). Digital image processing and development of machine learning models for the discrimination of corneal pathology: An experimental model. *Photonics, 8*(4), 118. https://doi.org/10.3390/photonics8040118

Comisión Federal para la Protección contra Riesgos Sanitarios. (2025). *Guía de requisitos para el trámite de registro sanitario de dispositivos médicos mediante acuerdos de equivalencia y vía regulatoria abreviada.* Secretaría de Salud, Gobierno de México. https://www.gob.mx/cofepris

Diario Oficial de la Federación. (2012, octubre 15). *Norma Oficial Mexicana NOM-004-SSA3-2012, Del expediente clínico.* Secretaría de Salud. https://dof.gob.mx/nota_detalle.php?codigo=5272787&fecha=15/10/2012

Google Cloud. (2026). *Vertex AI pricing and Medical Imaging Suite documentation.* Google LLC. Retrieved June 19, 2026, from https://cloud.google.com/vertex-ai/pricing

Marmor, M. F., Kellner, U., Lai, T. Y. Y., Melles, R. B., & Mieler, W. F. (2016). Recommendations on screening for chloroquine and hydroxychloroquine retinopathy (2016 revision). *Ophthalmology, 123*(6), 1386–1394. https://doi.org/10.1016/j.ophtha.2016.01.058

Pedregosa, F., Varoquaux, G., Gramfort, A., Michel, V., Thirion, B., Grisel, O., … Duchesnay, É. (2011). Scikit-learn: Machine learning in Python. *Journal of Machine Learning Research, 12*, 2825–2830.

Sandler, M., Howard, A., Zhu, M., Zhmoginov, A., & Chen, L.-C. (2018). MobileNetV2: Inverted residuals and linear bottlenecks. In *Proceedings of the IEEE Conference on Computer Vision and Pattern Recognition* (pp. 4510–4520). IEEE. https://doi.org/10.1109/CVPR.2018.00474

Selvaraju, R. R., Cogswell, M., Das, A., Vedantam, R., Parikh, D., & Batra, D. (2017). Grad-CAM: Visual explanations from deep networks via gradient-based localization. In *Proceedings of the IEEE International Conference on Computer Vision* (pp. 618–626). IEEE. https://doi.org/10.1109/ICCV.2017.74

Simonyan, K., & Zisserman, A. (2015). Very deep convolutional networks for large-scale image recognition. In *Proceedings of the 3rd International Conference on Learning Representations (ICLR).*

Studer, S., Bui, T. B., Drescher, C., Hanuschkin, A., Winkler, L., Peters, S., & Müller, K.-R. (2021). Towards CRISP-ML(Q): A machine learning process model with quality assurance methodology. *Machine Learning and Knowledge Extraction, 3*(2), 392–413. https://doi.org/10.3390/make3020020

Tan, M., & Le, Q. (2019). EfficientNet: Rethinking model scaling for convolutional neural networks. In *Proceedings of the 36th International Conference on Machine Learning* (PMLR 97, pp. 6105–6114).

van Velthoven, M. E., van der Linden, M. H., de Smet, M. D., Faber, D. J., & Verbraak, F. D. (2006). Influence of cataract on optical coherence tomography image quality and retinal thickness. *British Journal of Ophthalmology, 90*(10), 1259–1262. https://doi.org/10.1136/bjo.2004.097022